SurveyX: Academic Survey Automation via Large Language Models

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ABSTRACT

Large Language Models (LLMs) have demonstrated exceptional comprehension capabilities and a vast knowledge base, suggesting that LLMs can serve as efficient tools for automated survey generation. However, recent research related to automated survey generation remains constrained by some critical limitations like finite context window, lack of in-depth content discussion, and absence of systematic evaluation frameworks. Inspired by human writing processes, we propose SurveyX, an efficient and organized system for automated survey generation that decomposes the survey composing process into two phases: the Preparation and Generation phases. By innovatively introducing online reference retrieval, a pre-processing method called AttributeTree, and a re-polishing process, SurveyX significantly enhances the efficacy of survey composition. Experimental evaluation results show that SurveyX outperforms existing automated survey generation systems in content quality (0.259 improvement) and citation quality (1.76 enhancement), approaching human expert performance across multiple evaluation dimensions. Examples of surveys generated by SurveyX are available on our project website[1].

KEYWORDS

Automated Survey Generation, Literature Synthesis, Large Language Models, NLP

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1 INTRODUCTION

In recent years, computer science has advanced rapidly across various fields [2, 9, 24]. Statistics indicate that arXiv.org[2] receives about a thousand new papers daily. Figure 1 highlights a remarkable trend: over the past three years (2022-2024), the number of papers published on arXiv has increased substantially, rising from 186,339 to 285,174, representing a growth of over 50%. It is projected that this number will further climb to approximately 368,292 in 2025. However, the exponential growth of literature has made it increasingly challenging for researchers attempting to comprehend the technological evolution and developmental trajectories of specific subfields

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[1]http://www.surveyx.cn
[2]https://arxiv.org/

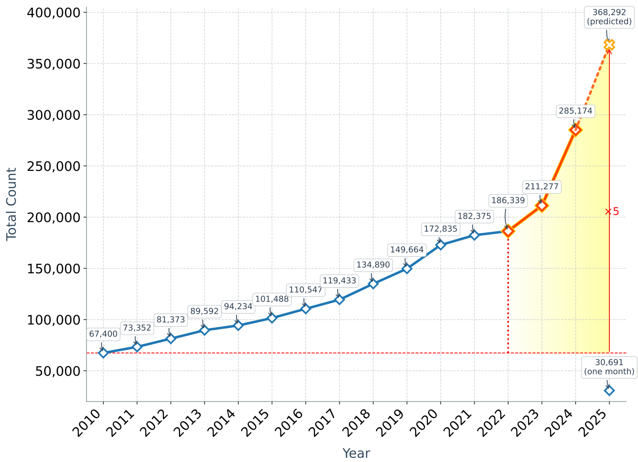

Figure 1: The number of papers received annually by the arXiv website from 2010 to 2025, with data sourced from our arXiv database. The projected number of submissions for 2025 is anticipated to be five times greater than that of 2010.

from the ground up [47]. Surveys are instrumental in elucidating the current state of research and the historical progression within a given topic [3, 45, 46]. However, the workload involved in manually writing these surveys is continually increasing, threatening the ability to maintain comprehensive coverage and high quality of these surveys. Against the backdrop of information overload, there is an urgent need to develop efficient systems for automated survey generation. The rise of Large Language Models (LLMs) has rendered automated survey generation a viable approach [13, 16, 37]. Trained on large-scale textual corpora, LLMs possess the capability to produce text that is both fluent and logically coherent [8, 23, 26, 34]. Despite this potential, leveraging LLMs for automated survey generation poses several challenges, which can be categorized into two levels:

1) Technical Challenges:

• LLMs primarily rely on their internal knowledge bases for text generation. However, effective survey composition requires comprehensive and accurate support from up-to-date references, whereas knowledge stored within LLMs may become outdated and occasionally provide incorrect reference information [11, 18, 27, 43]. This limitation causes negative effects on the academic rigor and credibility of the generated surveys. Consequently, relying solely on LLMs makes it difficult to generate high-quality surveys.

• Current mainstream LLMs encounter limitations due to their context window sizes [14, 19, 20]. For instance, GPT-4o has a context window of 128K tokens, while Claude 3.5 has 200K tokens. Crafting a comprehensive survey typically entails citing hundreds of references, each averaging around 10K tokens. This scale far exceeds the context window capacity of existing LLMs, posing a challenge to directly equipping them with all necessary references for generating high-quality surveys.

2) Application Challenges:

• The creation of surveys relies on a substantial number of references, necessitating timely content retrieval via online methods. However, there is currently a lack of effective tools that can efficiently procure large volumes of the latest and highly pertinent references, thus limiting the broader application of automated survey generation.

• Unlike traditional Natural Language Generation (NLG) tasks, the evaluation of surveys lacks unified metrics and standardized benchmarks [38]. The absence of evaluation frameworks hampers effective quality assessment of automatically generated surveys, thereby constraining their applicability in large-scale academic settings.

Existing similar works have provided some solutions to the aforementioned challenges, yet notable deficiencies persist, particularly in the following areas:

- Exisiting retrieval methodologies contain inherent limitations. In existing works, Wang et al. [38] only supports offline retrieval and cannot access the latest references, leading to a lack of timeliness.
- The methods for pre-processing references are often inadequate. In existing works, Wang et al. [38] only utilizes partial information from the references (titles and abstracts), overlooking a significant amount of crucial content.
- The generated surveys lack diversity in their expression. Existing methodologies are restricted to generating text-based surveys and fall short in including visual elements such as figures and tables, diminishing the readability of the results.

To address these deficiencies, we propose SurveyX, an efficient and well organized system for automated survey generation. SurveyX divides the composing process of surveys into two phases: the Preparation Phase and the Generation Phase. In the Preparation Phase, SurveyX employs retrieval algorithms to search and filter highly relevant references from the internet based on the given survey topic. It employs a reference pre-processing method, termed AttributeTree, to distill key information from the references, constructing a reference materials database for efficient retrieval through the Retrieval Augmented Generation (RAG) technique. In the Generation Phase, SurveyX utilizes the information obtained in the previous phase to sequentially generate the outline and main body of the survey, ensuring that the generated survey has a clear structure and accurate content. Additionally, tables and figures are incorporated to enrich the survey's presentation. Beyond the generation process, SurveyX extends the evaluation framework proposed by Wang et al. [38] by incorporating additional evaluation metrics, thus aiding subsequent related research.

To sum up, our main contributions are as follows:

- We propose an efficient reference retrieval algorithm capable of expanding keywords based on a given topic, substantially broadening the retrieval scope. Additionally, a 2-step filtering method is employed to eliminate papers with low relevance, leaving only high-quality references that comprehensively cover the topic.
- We design a reference pre-processing method called AttributeTree, which efficiently extracts key information from documents. This method significantly enhances the information density of reference materials, improving LLMs' comprehension and optimizing their context window usage.

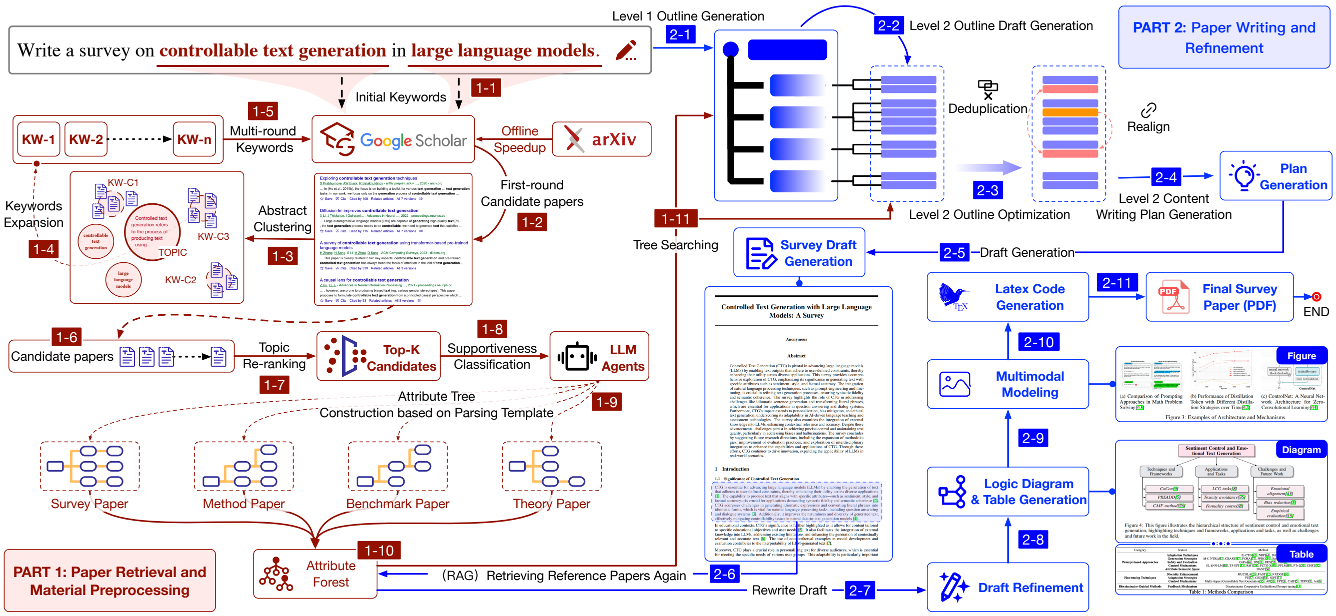

Figure 2: Pipeline of SurveyX.

- We introduce an outline generation method named Outline Optimization, which generates outlines based on hints and employs a "separate-then-reorganize" step to eliminate redundancy. This method results in outlines with more rigorous logic and clearer structure.
- We expand the expressive format of the generated survey to include figures and tables alongside text, enriching the presentation and improving readability.
- We augment the evaluation framework by introducing additional metrics to assess the quality of the generated surveys and retrieved references. The evaluation results demonstrate that the surveys generated by SurveyX outperform existing works across multiple metrics, closely aligning with human expert performance.

2 RELATED WORK

Long-form Text Generation. Although LLMs excel in traditional NLG tasks, generating long-form, well-structured, coherent, and logically organized text using LLMs still remains a persistent challenge [5, 17, 25, 28, 32, 36, 40]. To address this problem, some studies have sought to employ planning strategies. For instance, Tan et al. [31] proposed a method that first produces domain-specific content keywords, and then progressively refines them into complete passages in multiple stages. Similarly, Liang et al. [21] used a series of auxiliary training tasks to endow LLMs with the skills to plan and structure long-form documents before generating the final full article. Some other researches focus on the generation of long-form text in a specific format, such as Wikipedia articles, surveys, commentaries, etc [33, 35, 39, 42]. Shao et al. [30] developed STORM, a system that models the pre-writing stage by discovering diverse perspectives, simulating expert conversations, and curating information to create an outline, which is then used to generate a complete Wikipedia article. Wang et al. [38] introduced AotuSurvey, a framework for generating surveys consisting of initial retrieval, outline generation, parallel subsection drafting, integration, and rigorous evaluation. In comparison, our work addresses existing shortcomings by enhancing the retrieval scenarios, optimizing the reference pre-processing methods, expanding the expressive forms of the generated surveys, and improving overall quality.

Retrieval Augmented Generation (RAG). RAG technique is used to help LLMs access external knowledge, thereby enhancing their text generation capabilities [6, 7, 10, 12]. This technique has proven especially useful in tasks that require up-to-date or domain-specific information, such as QA (Question Answering) [1, 15], Dialog Generation [22, 29], Reasoning [4, 44], etc. Beyond traditional NLG tasks, RAG is also used for long-form text generation, as this task often requires handling extensive external knowledge [30, 38, 44]. Compared to previous work that directly used the raw text of references as the retrieval data source, our work significantly improves retrieval efficiency and context window utilization by converting it into attribute trees.

3 METHODOLOGY

In this section, we delineate the pipeline of SurveyX, which includes two main phases: Preparation Phase (section 3.1) and Generation Phase (section 3.2). The primary task of the Preparation Phase is to collect the related references needed for composing the survey and pre-processing them to construct a reference materials database for retrieval using RAG technique. The Generation Phase undertakes the creation of both the outline and main body of the survey based on these materials. This phase also involves the post-processing refinement of the initial draft to enhance its overall quality and augment its expressive presentation. An overview of this system is depicted in Figure 2.

3.1 Preparation Phase

Preparation Phase consists of two stages: the references acquisition stage and the references pre-processing stage.

3.1.1 References Acquisition. Considering the current shortage of effective tools for efficiently acquiring a large volume of highly relevant references, we independently develop a module for acquiring references. This module comprises two separate components: **retrieval data source** and **retrieval algorithms**.

The **retrieval data source** includes both offline and online data sources. The offline data source consists of 2,632,189[3] available papers downloaded from the arXiv.org, with new papers added daily. The online data source is a self-developed crawler system based on Google Scholar. Using the offline data source facilitates quick access to existing references, thereby saving time, while the online data source enables the acquisition of the latest and multi-source references. The combination of both ensures efficiency and timeliness.

The **retrieval algorithm** is divided into two steps: reference recall and reference filter (steps 1-1~1-6 and steps 1-7~1-8 in Figure 2). (1) The goal of the reference recall step is to avoid missing any references related to the topic as much as possible. (2) The paper filter step aims to filter out references unrelated to the topic as much as possible. To achieve (1), we design a method called the Keyword Expansion Algorithm. The pseudo code can be found at Algorithm 1. The algorithm initializes the keyword pool with an initial keyword K_0. After searching references by newly added keywords, it performs semantic clustering on the abstracts of all retrieved references *Doc* and summarizes keywords for each cluster C_i. The obtained keywords K_c are then semantically compared with the existing keyword pool K_{pool} and the topic T, the most appropriate keyword k^* based on semantic distance is added to the keyword pool. This process is repeated until the number of *Doc* reaches the threshold θ (set as 1000).

Algorithm 1 Keyword Expansion Algorithm

Input: initial keywords set K_0, topic T
Output: retrieved documents Doc
keyword pool: $K_{\text{pool}} \leftarrow K_0$
while LengthOf(Doc) $< \theta$ **do**
  $k \leftarrow$ newly added keywords in K_{pool}
  $Doc \leftarrow Doc + \text{Search}(k)$
  $n \leftarrow \text{LengthOf}(K_{\text{pool}})$
  $C \leftarrow \text{Cluster}(Doc, n+1)$
  $K_c \leftarrow \text{ExtractKeywords}(C)$
  $Dist \leftarrow \text{CalcDist}(K_c, K_{\text{pool}}, T, \text{Weights})$
  $k^* \leftarrow \text{Select}(Dist, K_{\text{pool}})$
  $K_{\text{pool}} \leftarrow K_{\text{pool}} + k^*$
end while
Return: Doc

Weights are the semantic distance weights. Considering that all keywords should more closely revolve around the topic, we double the semantic distance weight of the topic. The *Select* function, when choosing candidate words, assumes that the best candidate word should have the smallest possible average distance to the existing keywords while also having the maximum furthest distance. Specifically:

$$k^* = \arg\min_{k_c \in K_C} \big(R_1(k_c) + R_2(k_c)\big) \tag{1}$$

R_1 and R_2 represent different ranking calculation methods. Their calculation methods are as follows:

$$R_1(k_c) = \text{rank}_{K_c}\left(\frac{1}{|K_{pool}|}\sum_{k_e \in K_{pool}} \text{cos\_sim}(E(k_c), E(k_e))\right) \tag{2}$$

$$R_2(k_c) = \text{rank}_{K_c}\left(-\max_{k_e \in K_{pool}} \text{cos\_sim}(E(k_c), E(k_e))\right) \tag{3}$$

Here, rank$(\cdot)$ refers to the Rank Function, and $E(\cdot)$ represents the embedding model. To achieve (2), we propose a 2-step filtration algorithm. The first step uses an embedding model to calculate the semantic relevance between the topic and abstracts of references, selecting the Top-K references most relevant to the topic, which serves as coarse-grained filtration. The second stage uses LLMs for more precise semantic filtering, serving as fine-grained filtration. This algorithm ensures that the selected papers maintain a high degree of relevance to the research topic, thereby enhancing the quality of the survey.

3.1.2 References Pre-processing. After obtaining the references, they need to be preprocessed for use in subsequent stages. A naive and common pre-processing method is to directly provide the full text of the references to LLMs for generating the surveys. However, we believe this approach suffers from low context window utilization and inefficient extraction of key information. We have noticed that before composing a survey, people often categorize the necessary references and organize all potentially useful information. Based on this observation, we design a reference pre-processing method called AttributeTree. Specifically, we designed different attribute tree templates for different types of references in advance. Using these templates, we can efficiently extract key information from the references. The attribute trees of all references are combined into an attribute forest, which represents all the reference materials required for composing a survey (steps 1-9~1-10 in Figure 2). Examples of attribute tree templates can be found in the Appendix B. This method significantly increases the information density of the reference materials and efficiently utilizes the context window of LLMs, thus laying a solid foundation for composing high-quality surveys.

3.2 Generation Phase

After obtaining all the reference materials needed for writing the survey, the generation phase begins. This phase is divided into three stages: **outline generation**, **main body generation**, and **post refinement**.

3.2.1 Outline Generation. Outline generation is the most crucial stage in the entire survey generation process. A well-structured, logically organized, and focused outline ensures a comprehensive, cohesive, and informative survey. Our preliminary experiments indicate that, when generating the primary outline, utilizing only

[3]Data as of February 10, 2025.

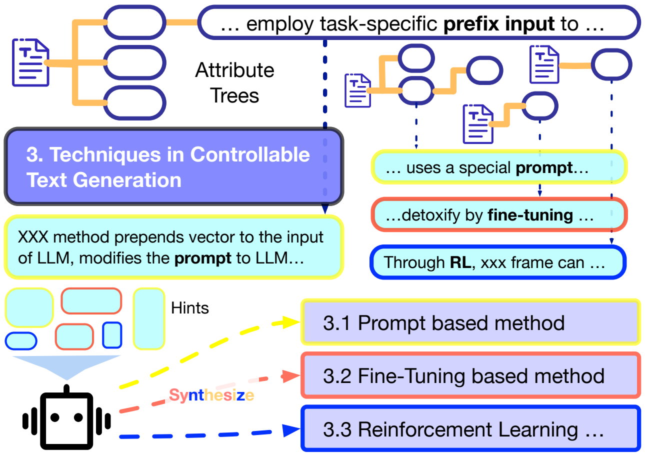

Figure 3: An example of generating secondary outlines. LLMs first generate hints based on the attribute tree to guide the generating of the secondary outline. Then, by synthesizing all hints, LLMs identify the most suitable entry points to determine the segmentation strategy and generate the secondary outline.

the internal knowledge of LLMs is sufficient to produce a human-level primary outline. In contrast, generating the secondary outline is more challenging, and relying solely on the internal knowledge of LLMs is inadequate. We observed that humans typically categorize or summarize references to guide the writing of the survey. Inspired by this, we designed an outline generation method named Outline Optimization (steps 2-1~2-3 in Figure 2). This method consists of two steps.

For the first step, this method initially lets LLMs generate hints corresponding to the secondary outline based on the attribute tree of each reference. A hint can be considered as a form of guiding content that assists LLMs in better understanding the key information a particular reference can provide in constructing a systematic framework[4]. Then, LLMs sequentially generate the secondary outline based on these hints. An example of this step is shown in Figure 3. Through this step, LLMs can more accurately identify the commonalities and differences among various references and more efficiently synthesize and organize these references, thereby improving the generation of secondary outlines.

For the second step, this method begins by separating all secondary outlines from the primary outlines, retaining only their headings to facilitate deduplication by the LLMs. Then, use LLMs to reorganize the deduplicated secondary outlines into the primary outline. Through this step, the method effectively addresses the redundancy present in the generated secondary outlines.

With the help of the outline optimization method, we can generate survey outlines with logical rigor and clear structure.

3.2.2 Content Generation. During the content generation stage, inspired by the human writing process for surveys, we continue to use the hint-based method employed during the outline generation stage to enhance the depth of the generated content (steps 2-4~2-5 in Figure 2). Unlike before, the hints used here are derived from the secondary outline, aiming to guide the generation of the main body content. LLMs will generate the main body content on a subsection[5] basis using these hints and the outline produced in the previous stage. Meanwhile, when the LLM is writing a specific subsection, it can view the content of other subsections. This ensures that the content currently being generated remains consistent with the content already generated to some extent.

3.2.3 Post Refinement. In the human process of writing surveys, revising and polishing the draft to enhance its quality is common. Based on this, we design a post refinement stage, which polishes the initial draft with two main goals: (1) to improve the quality of the content, including citation quality, textual fluency, and consistency of expression; (2) to add figures and tables to enrich the presentation of the survey. These two objectives are achieved by an RAG-based rewriting module and a graph and table generation module, respectively.

RAG-based Rewriting Module. Even though strategies were used in the content generation stage to ensure content consistency in the generated survey, these efforts are not completely sufficient. Additionally, the accuracy of citations in the generated survey has not been verified. To address these issues, we design a RAG-based rewriting module to revise the survey content. This module first uses the paragraphs from the initial draft as queries to retrieve reference materials from the attribute forest. Subsequently, it constructs a prompt based on these materials to rewrite the paragraph using LLMs (steps 2-6~2-7 in Figure 2). The rewriting process involves two aspects: first, removing irrelevant citations and adding highly relevant ones to the paragraph, and second, polishing the paragraph by considering its context. This module not only significantly enhances the citation quality but also ensures the content consistency of the generated survey.

Figure and Table Generation Module. Surveys that consist solely of text often lack expression diversity, which can limit their effectiveness in conveying information. To address this issue, we designed a figure and table generation module (steps 2-8~2-11 in Figure 2). Inspired by Napkin[6], We construct several information extraction templates, each corresponding to a specific figure or table generation template (the generation templates are scripts). Based on the information extraction templates, we use LLMs to extract the necessary information from the attribute tree of the references for generating figures or tables. Subsequently, the generation templates automatically construct the corresponding figures or tables based on the extracted information. Additionally, on a subchapter basis, we leverage Multimodal Large Language Models (MLLMs) combined with context to retrieve figures from references. If a figure effectively supports the content of a subsection, we incorporate it to enhance clarity and expressiveness.

Through the aforementioned modules, we have constructed a comprehensive and systematic post refinement stage that significantly enhances the overall quality of the generated survey while expanding its expressive format.

[4]"systematic framework" refers to the framework proposed in a survey for organizing and synthesizing all the references.

[5]One secondary outline corresponds to one subsection.

[6]https://app.napkin.ai/

4 EXPERIMENTS

4.1 Evaluation Metrics

Automatic evaluation. In terms of content quality evaluation, we expanded the evaluation dimensions presented in [38] by incorporating synthesis and critical analysis into the evaluation metrics. Synthesis evaluates the ability to interconnect disparate studies, identify overarching patterns or contradictions, and construct a cohesive intellectual framework beyond individual summaries, while critical analysis examines the depth of critique applied to existing studies, including the identification of methodological limitations, theoretical inconsistencies, and research gaps.

For the evaluation of citation quality, we adopted the Citation Recall and Citation Precision metrics proposed by [38]. The former evaluates whether each statement in the generated text is fully supported by the cited references, while the latter evaluates for the presence of irrelevant citations. In addition, we introduced the F1 Score metric to provide a more comprehensive evaluation of citation quality.

Regarding the evaluation of reference relevance, we identify an existing gap in the literature, as this aspect has not been addressed in previous studies. To fill this void, we propose a novel set of metrics specifically designed to assess the relevance of retrieved references. The details of this proposed metric set are as follows:

(1) IoU (Insertion over Union) This metric measures the similarity between machine-retrieved and human-retrieved references by calculating the degree of overlap between the two. The calculation method is as follows

$$IoU = \frac{Doc_{human} \cap Doc_{machine}}{Doc_{human} \cup Doc_{machine}}$$

(2) Relevance$_{semantic}$ (semantic-based reference relevance). This metric is based on an embedding model, calculating the cosine similarity between the embeddings of the retrieved references and the topic to measure their relevance. The calculation method is as follows:

$$Relevance_{semantic} = \frac{1}{|Doc|} \sum_{d \in Doc} \text{cos\_sim}(E(d), E(topic))$$

(3) Relevance$_{LLM}$ (LLM-based reference relevance). This metric is designed by constructing prompts to leverage LLMs to directly evaluate the relevance of the retrieved references to the topic. The calculation method is as follows:

$$Relevance_{LLM} = \frac{1}{|Doc|} \sum_{d \in Doc} \mathbb{I}_{\text{relevant}}(LLM(Prompt(d, topic)))$$

The prompts used in the evaluation are presented in the Appendix. A.3

Human evaluation. In addition to automated evaluation, we also incorporate human evaluation. Human evaluation not only verifies the reliability of automated evaluation but also addresses complex factors such as contextual information and implicit logic, which are difficult for automated evaluation to capture. The evaluation criteria used for human evaluation are the same as that for automated evaluation. Considering the costs, we employ human evaluation only for content quality evaluation. The details of the human evaluation are presented in the Appendix. A.2

4.2 Experiment Settings

Implementation. During the retrieval stage and the evaluation stage, we selected bge-base-en-v1.5 [41] as the embedding model. Throughout the entire process, we use GPT4o as our LLM agent.[7].

Baselines. We employ the following methods as our baselines:

Human. Human-written surveys collected from arxiv.org. Detailed information can be found at www.surveyx.cn.

Navie RAG: We used the same references as SurveyX to provide abstracts for the LLM to guide the generation of the survey. The prompt used by Naive RAG is presented in the Appendix A.4.

AutoSurvey: An automated survey generation system proposed by [38]. It divides the survey generation process into four stages: Initial Retrieval and Outline Generation, Subsection Drafting, Integration and Refinement, and Rigorous Evaluation and Iteration. In our experiments, we employed its 64k version.

Test Cases. We adopted the 20 topics mentioned by [38] to generate the corresponding surveys for comparison. Details of these topics can be found in the Table 4.

Ablation. To evaluate the impact of different modules or methods on the performance of SurveyX, we designed the following ablation setting:

(1) For the retrieval algorithms (component) within the retrieval module of the references acquisition stage, we directly use the initial keywords for retrieval, bypassing the Keyword Expansion retrieval algorithm.

(2) For the AttributeTree method in the references pre-processing stage, we replace the originally generated attribute trees with the full text of the references.

(3) For the outline optimization method in the outline generation stage, we design prompts to let the LLM generate the entire outline in one step.

(4) For the RAG-based rewriting module in the post refinment stage, we eliminate the entire RAG-based rewriting module.

[7]Specifically, we use gpt-4o-2024-08-06

Table 1: Content quality evaluation results of naive RAG, Autosurvey, SurveyX and Human writing. All LLM-Agent is GPT-4o.

Model	Coverage	Structure	Relevance	Synthesis	Critical Analysis	Avg	Recall	Precision	F1
naive RAG	4.40	3.66	4.66	3.82	2.82	3.872	68.79	61.97	65.20
AutoSurvey	4.73	4.33	4.86	4.00	3.73	4.331	82.25	77.41	79.76
SurveyX	**4.95**	**4.91**	**4.94**	**4.10**	**4.05**	**4.590**	**85.23**	**78.12**	**81.52**
Human	5.00	4.95	5.00	4.44	4.38	4.754	86.33	77.78	81.83

4.3 Experiment Results and Analysis

Table 3: Evaluation results of reference relevance.

Model	IoU	$Relevance_{semantic}$	$Relevance_{LLM}$
Human	1	0.4455	0.9485
SurveyX	0.55	0.4226	0.7689

Main results. The results for content quality evaluation, citation quality evaluation, and reference relevance evaluation are shown in Table 1 and Table 3. Key findings are as follows:

(1) The experimental results for content quality evaluation indicate that SurveyX performs exceptionally well across all metrics, particularly in Coverage (4.95), Structure (4.91), and Relevance (4.94), closely approaching the performance of human experts. This demonstrates that the surveys generated by SurveyX not only comprehensively cover both the core and peripheral content of the given topic but also ensure logical and coherent outlines while maintaining high relevance. Compared to naive RAG and AutoSurvey, SurveyX shows a clear advantage across all metrics, with significant improvements in Structure (4.91) and Critical Analysis (4.05). This indicates that the surveys generated by SurveyX have clearer and more coherent outlines and greater depth in content. Additionally, SurveyX leads existing automated survey generation systems with an average score of 4.590, proving its outstanding performance in survey generation.

(2) The results of the citation quality experiments show that SurveyX outperforms existing automated survey generation systems and closely approaches the performance of human experts in Citation Recall (85.23), Citation Precision (78.12), and F1 Score (81.52). Notably, in the Precision metric (78.12), SurveyX even slightly surpasses human experts. This indicates that SurveyX can significantly enhance the comprehensiveness and accuracy of citations in its generated surveys, ensuring that all statements are well-supported by literature while effectively reducing irrelevant citations, thereby increasing the reliability and credibility of the generated surveys. These experimental results also indirectly demonstrate that the RAG-based rewriting component used in the post refinment stage of SurveyX effectively improves the citation quality of the generated surveys.

(3) The results of the reference relevance experiments indicate that SurveyX approaches the performance of human experts in the $Relevance_{semantic}$ metric. However, there is a certain gap compared to human experts in the IoU and $Relevance_{LLM}$ metrics. Nonetheless, as the first automated survey generation system in this field with comprehensive online reference retrieval capabilities, SurveyX demonstrates its potential and application prospects, laying the foundation for future research.

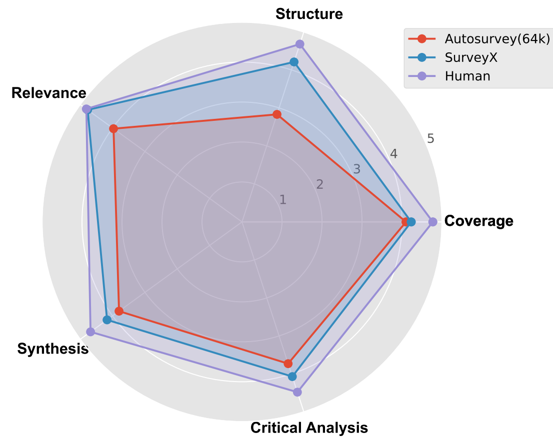

Figure 4: Human evaluation results.

Human evaluation results. The results of the human evaluation are shown in Figure 4. The results show that SurveyX outperforms AutoSurvey across all metrics and is closer to the performance of human experts. This aligns with the results of the automated evaluation, which to some extent supports the validity of our automated evaluation method. Additionally, compared to the automated evaluation results, the scores from human evaluation are generally lower, especially in the Structure metric. This reflects the higher standards of human evaluators for the survey in areas such as structural coherence, logical consistency, content depth, etc.

Ablation results. The results of the ablation experiments are shown in Table 2. The experimental results indicate that each module of SurveyX plays a crucial role. Key findings are as follows:

(1) After ablating the retrieval algorithm, the Coverage (4.95→4.74) and Relevance (4.94→4.79) metrics showed the most significant decreases. This indicates that the improvement in reference quality brought by the retrieval algorithm is crucial for ensuring comprehensive content coverage and high topic relevance in the generated surveys.

(2) After ablating the AttributeTree method, the Structure (4.91→4.08), Synthesis (4.10→3.88), and Critical Analysis (4.05→3.93) metrics showed significant declines. Besides, the Recall (85.23→60.09),

Table 2: Ablation study results for SurveyX with different components removed. Data with significant declines are indicated by underlines.

Ablation Object	Coverage	Structure	Relevance	Synthesis	Critical Analysis	Avg	Recall	Precision	F1
Retrieval Algorithm	4.74	4.88	4.79	3.98	4.02	4.48	78.88	73.34	76.01
AttributeTree Method	4.84	4.08	4.89	3.88	3.93	4.32	60.09	56.49	58.23
Outline Optimization Method	4.90	3.80	4.91	3.98	4.02	4.32	85.1	77.13	80.92
RAG-based Rewriting Module	4.92	4.89	4.93	4.00	4.00	4.55	55.37	54.95	55.16
No Ablation	**4.95**	**4.91**	**4.94**	**4.10**	**4.05**	**4.590**	**85.23**	**78.12**	**81.52**

Precision (78.12→56.49), and F1 score (81.52→58.23) metrics also decreased noticeably. This indicates that converting the content of the reference into an attribute tree enables the LLM to understand the core information better, thereby improving its ability to organize outlines, integrate information from different references, conduct in-depth analysis, and add references more accurately.

(3) After ablating the outline optimization method, the Structure metric (4.91→3.80) showed the most significant decline. This indicates that this method effectively enhances the logical coherence and structural clarity of the generated survey outline.

(4) After ablating the RAG-based rewriting module, the Recall (85.23→55.37), Precision (78.12→54.95), and F1 score (81.52→55.16) metrics declined significantly. This indicates that the RAG-based rewriting module can considerably enhance the citation quality of the generated survey by more accurately adding relevant citations and removing irrelevant ones.

5 CONCLUSION

In this paper, we propose SurveyX, an innovative automated survey generation system. SurveyX effectively addresses issues in LLM-based automated survey generation, such as context window limitations and internal knowledge constraints. Moreover, SurveyX also overcomes shortcomings of existing automatic generation systems, including a lack of diversity in survey expression, inadequate reference pre-processing methods, and limitations in retrieval approaches. Experimental results demonstrate that SurveyX significantly outperforms existing systems in both content and citation quality, approaching the level of human experts. This achievement indicates that SurveyX can serve as a reliable and efficient tool for automated survey generation, providing valuable assistance to researchers.

In the future, we consider improving our system in the following areas:

- Optimizing the retrieval algorithm to achieve retrieval performance comparable to human levels.
- Expanding the methods for generating figures and tables.
- Enhancing the organization of surveys by further refining the composing approach based on the attribute tree.

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A EVALUATION DETAILS

A.1 Survey Topics

Table 4 presents the 20 survey topics used in the evaluation experiment, along with the titles of the corresponding survey written by human experts.

A.2 Details of human evaluation

For the human evaluation, six PhD students with experience in writing LLM-related surveys were invited. The platform used for evaluation is Label Studio[8]. The final score was derived by averaging the scores given by all evaluators.

A.3 Evaluation Prompt

Prompts used for content evaluation are presented in figure 5, 6, 7, 8, 9. Prompts used for citation quality are presented in figure 10. Prompts used for reference papers are presented in figure 11.

A.4 Naive RAG Prompt

Prompts used for Naive RAG method are presented in figure 12

B ATTRIBUTE TREE TEMPLATES

Attribute tree templates are shown in figure 13, 14, 15, 16.

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[8]https://labelstud.io/

Table 4: Survey Topics

Topic	Survey Title
In-context Learning	A survey for in-context learning
LLMs for Recommendation	A Survey on Large Language Models for Recommendation
LLM-Generated Texts Detection	A Survey of Detecting LLM-Generated Texts
Explainability for LLMs	Explainability for Large Language Models
Evaluation of LLMs	A Survey on Evaluation of Large Language Models
LLMs-based Agents	A Survey on Large Language Model based Autonomous Agents
LLMs in Medicine	A Survey of Large Language Models in Medicine
Domain Specialization of LLMs	Domain Specialization as the Key to Make Large Language Models Disruptive
Challenges of LLMs in Education	Practical and Ethical Challenges of Large Language Models in Education
Alignment of LLMs	Aligning Large Language Models with Human
ChatGPT	A Survey on ChatGPT and Beyond
Instruction Tuning for LLMs	Instruction Tuning for Large Language Models
LLMs for Information Retrieval	Large Language Models for Information Retrieval
Safety in LLMs	Towards Safer Generative Language Models: Safety Risks, Evaluations, and Improvements
Chain of Thought	A Survey of Chain of Thought Reasoning
Hallucination in LLMs	A Survey on Hallucination in Large Language Models
Bias and Fairness in LLMs	Bias and Fairness in Large Language Models
Large Multi-Modal Language Models	Large-scale Multi-Modal Pre-trained Models
Acceleration for LLMs	A Survey on Model Compression and Acceleration for Pretrained Language Models
LLMs for Software Engineering	Large Language Models for Software Engineering

Evaluation prompt: content - coverage

Here is an academic survey about the topic "{topic}":

{content}

<instruction>
Please evaluate this survey about the topic {topic} based on the criterion above provided below, and give a score from 1 to 5 according to the score description:

Criterion Description: Coverage: Coverage assesses the extent to which the survey encapsulates all relevant aspects of the topic, ensuring comprehensive discussion on both central and peripheral topics.

Score 1 Description: The survey has very limited coverage, only touching on a small portion of the topic and lacking discussion on key areas.
Score 2 Description: The survey covers some parts of the topic but has noticeable omissions, with significant areas either underrepresented or missing.
Score 3 Description: The survey is generally comprehensive in coverage but still misses a few key points that are not fully discussed.
Score 4 Description: The survey covers most key areas of the topic comprehensively, with only very minor topics left out.
Score 5 Description: The survey comprehensively covers all key and peripheral topics, providing detailed discussions and extensive information.

Return the score without any other information:

Figure 5: Content coverage prompt for evaluation.

Evaluation prompt: content - structure

Here is an academic survey about the topic "{topic}":

{content}

<instruction>
Please evaluate this survey about the topic {topic} based on the criterion above provided below, and give a score from 1 to 5 according to the score description:

Criterion Description: Structure: Structure evaluates the logical organization and coherence of sections and subsections, ensuring that they are logically connected.

Score 1 Description: The survey lacks logic, with no clear connections between sections, making it difficult to understand the overall framework.
Score 2 Description: The survey has weak logical flow with some content arranged in a disordered or unreasonable manner.
Score 3 Description: The survey has a generally reasonable logical structure, with most content arranged orderly, though some links and transitions could be improved such as repeated subsections.
Score 4 Description: The survey has good logical consistency, with content well arranged and natural transitions, only slightly rigid in a few parts.
Score 5 Description: The survey is tightly structured and logically clear, with all sections and content arranged most reasonably, and transitions between adajecent sections smooth without redundancy.

Return the score without any other information:

Figure 6: Content structure prompt for evaluation.

Evaluation prompt: content - relevance

Here is an academic survey about the topic "{topic}":

{content}

<instruction>
Please evaluate this survey about the topic {topic} based on the criterion above provided below, and give a score from 1 to 5 according to the score description:

Criterion Description: Relevance: Relevance measures how well the content of the survey aligns with the research topic and maintain a clear focus.

Score 1 Description: The content is outdated or unrelated to the field it purports to review, offering no alignment with the topic
Score 2 Description: The survey is somewhat on topic but with several digressions; the core subject is evident but not consistently adhered to.
Score 3 Description: The survey is generally on topic, despite a few unrelated details.
Score 4 Description: The survey is mostly on topic and focused; the narrative has a consistent relevance to the core subject with infrequent digressions.
Score 5 Description: The survey is exceptionally focused and entirely on topic; the article is tightly centered on the subject, with every piece of information contributing to a comprehensive understanding of the topic.

Return the score without any other information:

Figure 7: Content relevance prompt for evaluation.

Evaluation prompt: content - synthesis

Here is an academic survey about the topic "{topic}":

{content}

<instruction>
Please evaluate this survey about the topic {topic} based on the criterion above provided below, and give a score from 1 to 5 according to the score description:

Criterion Description: Synthesis: Synthesis evaluates the ability to interconnect disparate studies, identify overarching patterns or contradictions, and construct a cohesive intellectual framework beyond individual summaries.

Score 1 Description: The survey is purely a collection of isolated study summaries with no attempt to connect ideas or identify broader trends.
Score 2 Description: The survey occasionally links studies but fails to synthesize them into meaningful patterns; connections are superficial.
Score 3 Description: The survey identifies some thematic relationships between studies but lacks a unified framework to explain their significance.
Score 4 Description: The survey synthesizes most studies into coherent themes or debates, though some connections remain underdeveloped.
Score 5 Description: The survey masterfully integrates studies into a novel framework, revealing latent trends, resolving contradictions, and proposing paradigm-shifting perspectives.

Return the score without any other information:

Figure 8: Content synthesis prompt for evaluation.

Evaluation prompt: content - critical analysis

Here is an academic survey about the topic "{topic}":

{content}

<instruction>
Please evaluate this survey about the topic {topic} based on the criterion above provided below, and give a score from 1 to 5 according to the score description:

Criterion Description: Critical Analysis: Critical Analysis examines the depth of critique applied to existing studies, including identification of methodological limitations, theoretical inconsistencies, and research gaps.

Score 1 Description: The survey merely lists existing studies without any analytical commentary or critique.
Score 2 Description: The survey occasionally mentions limitations of studies but lacks systematic analysis or synthesis of gaps.
Score 3 Description: The survey provides sporadic critical evaluations of some studies, but the critique is shallow or inconsistent.
Score 4 Description: The survey systematically critiques most key studies and identifies research gaps, though some areas lack depth.
Score 5 Description: The survey demonstrates rigorous critical analysis of methodologies and theories, clearly maps research frontiers, and proposes novel directions based on synthesized gaps.

Return the score without any other information:

Figure 9: Content critical analysis prompt for evaluation.

Evaluation prompt: citation

Claim:
{claim}

Source:
{source}

Claim:
{claim}

Is the Claim faithful to the Source?
A Claim is faithful to the Source if the core part in the Claim can be supported by the Source.\n
Only reply with 'Yes' or 'No':

Figure 10: Citation prompt for evaluation.

Evaluation prompt: reference paper relevance

I will give you a paper abstract and a topic, you decide if this paper is relevant to this topic.
Only return "yes" or "no".
Here is the paper abstract:
{abstract}
Here is the topic:
{topic}
So, is this paper relevant to this topic?
Only output "yes" or "no".

Figure 11: Judge relevance for reference paper.

Prompt for Naive RAG

- Role: Academic Review Paper Author and Research Synthesizer
- Background: The user has gathered numerous abstracts and has a clear topic and title for the academic review paper. The goal is to synthesize these materials into a coherent academic review paper in Markdown format. The topic is {topic}, the title is {title}.
- Profile: As an Academic Review Paper Author, you possess a deep understanding of academic writing standards, research synthesis techniques, and the ability to articulate complex ideas clearly and concisely. You are adept at organizing information logically and presenting it in a structured manner suitable for academic discourse.
- Skills: Proficiency in academic writing, literature synthesis, critical analysis, and the ability to format content in Markdown for clarity and readability.
- Goals: To create a well-structured academic review paper that integrates the information from the provided abstracts, adheres to academic standards, and is formatted in Markdown.
- Constrains: The paper must maintain academic integrity, ensure that all sources are properly cited. The content should be original and synthesized from the abstracts provided, not merely a summary.
- OutputFormat: Markdown formatted academic review paper with sections such as Introduction, Literature Review, Discussion, and Conclusion.
- Workflow:
1. Analyze the provided abstracts to identify key themes and findings.
2. Organize the information into a logical structure that aligns with the topic and title of the review paper.
3. Write the paper in Markdown format, ensuring that each section is clearly delineated and that the content flows coherently.

The abstracts is:
{abstracts}

Now write the academic survey:

Figure 12: Prompts for Naive RAG.

Method Paper

1. ***Background***: *problem background, including previous methods, the need of new breakthrough.*
2. ***Problem***
   a. *definition: specific description of problem.*
   b. *key obstacle: main difficulty, main challenge.*
3. ***Idea***
   a. *intuition: idea was inspired by what.*
   b. *opinion: what's the idea*
   c. *innovation: what's the main difference compared to previous method, or where is the primary improvement.*
4. ***Method***
   a. *method definition: given the problem, what's the definiton of method.*
   b. *method description: in one sentence, describe the method.*
   c. *method steps: procedures of method.*
   d. *principle: why this method is effective.*
5. ***Experiments***
   a. *experiments setting: including dataset, baseline and so on.*
   b. *experiments progress: specific evaluation steps*
6. ***Conclusion***: *what's the conclusion of experiments/paper.*
7. ***Discussison***
   a. *advantage: what's the advantages of this paper.*
   b. *limitation: what's the disadvantages of this paper.*
   c. *future work: based on the adv and disadv, what and where can be improved in the future.*
8. ***Other Info***: *Is there any other info not mentioned above?*

Figure 13: Method paper attribute tree.

Benchmark Paper

1. ***Background***: *problem background, including previous methods, the need of new breakthrough.*
2. ***Problem***
   a. *definition: specific description of problem.*
   b. *key obstacle: main difficulty, main challenge.*
3. ***Idea***
   a. *intuition: idea was inspired by what.*
   b. *opinion: what's the idea*
   c. *innovation: what's the main difference compared to previous method, or where is the primary improvement.*
4. ***Dataset***
   a. *source: how this dataset generated.*
   b. *desc: a description of dataset, eg. scale, accessibility, diversity, quality and so on.*
   c. *content: the specific content of dataset.*
5. ***Metrics***
   a. *aspect: what aspects of the model performance are being measured.*
   b. *principle: the underlying principles of the metrics used.*
   c. *procedure: how to evaluate performance of a model?*
6. ***Experiments***
   a. *model: which models used in the experiments.*
   b. *procedure: how the experiment conducted, also as the experiments settings.*
   c. *result: what's the result of experiments. How models perform on this benchmark, if is statistically significant.*
   d. *variability: how was the variability in the results accounted for?*
7. ***Conclusion***: *what's the conclusion of experiments/paper.*
8. ***Discussion***
   a. *advantage: what's the advantages of this paper.*
   b. *limitation: what's the disadvantages of this paper.*
   c. *future work: based on the adv and disadv, what and where can be improved in the future.*
9. ***Other Info***: *Is there any other info not mentioned above?*

Figure 14: Benchmark paper attribute tree.

Theory Paper
1. ***Background***: *problem background, including previous methods, the need of new breakthrough.*
2. ***Problem***
a. *definition: specific description of problem.*
b. *key obstacle: main difficulty, main challenge.*
3. ***Idea***
a. *intuition: idea was inspired by what.*
b. *opinion: what's the idea*
c. *innovation: what's the main difference compared to previous method, or where is the primary improvement.*
4. ***Theory***
a. *perspective: the perspective of theory and the architecture of the perspective.*
b. *opinion: the view or assumption about a problem.*
c. *proof: the proof or derivation of a theory.*
5. ***Experiments***
a. *experiments setting: including dataset, baseline and so on.*
b. *experiments progress: specific evaluation steps*
6. ***Conclusion***: *what's the conclusion of experiments/paper.*
7. ***Discussison***
a. *advantage: what's the advantages of this paper.*
b. *limitation: what's the disadvantages of this paper.*
c. *future work: based on the adv and disadv, what and where can be improved in the future.*
8. ***Other Info***: *Is there any other info not mentioned above?*

Figure 15: Theory paper attribute tree.

Survey Paper
1. Background:
a. *purpose: the purpose of this survey.*
b. *scope: topics covered and what not covered by this survey.*
2. Problem
a. *definition: specific description of problem.*
b. *key obstacle: main difficulty, main challenge.*
3. Architecture:
a. *perspective: the new view this survey proposed, which could summarize each method into a fields/stages.*
b. *fields/stages: what fields or stages the survey summarize the current methods to?*
4. Conclusion: *what's the conclusion of experiments/paper.*
a. *comparisons: comparisons between each research or methods.*
b. *results: the main conclusion or discovery.*
5. Discussion
a. *advantage: what's the advantages of existing research.*
b. *limitation: what's the disadvantages of existing research.*
c. *gaps: gaps existed in the current researches.*
d. *future work/trends: based on the adv and disadv, what and where can be improved in the future.*
6. Other Info: *is there any other info not mentioned above? List them in json format.*

Figure 16: Survey paper attribute tree.